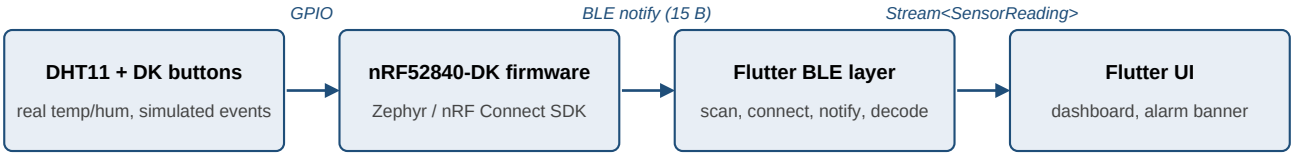
SowGuard - Lab Report 1: Work Split

IoT Laboratory (UAB) - Device + Edge over BLE - Deadline: 30 Oct 2026 (Virtual Campus)

1. Scope of this delivery

Report 1 covers only the **Device** (nRF52840-DK programmed to send data over BLE) and the **Edge** (Flutter app on Android receiving and showing that data). Cloud, database and ML are **not** part of this delivery. The lab prototype is a simplified version of the design: one single node, BLE instead of Thread, and simulated data wherever we have no real sensor (allowed by the professor). We keep the code simple because we will be asked about it.

2. Prototype overview



Signal	Source in the lab	How
Temperature / humidity	**Real**: DFR0067 (DHT11)	Zephyr sensor driver, read every 2 s
Water flow	**Simulated** in firmware	Base value + noise; a slow drop can be triggered for the demo
Sow lying down (accelerometer in design)	**Simulated**: DK Button 1	GPIO interrupt toggles posture
Piglet scream (microphone in design)	**Simulated**: DK Button 2	Lying + scream within N s sets the crush alarm flag. No audio is sent
Status	DK LEDs	LED1 = connected, LED2 = alarm

3. Shared contract (frozen by 23 Oct)

Everything below is the interface between the three members. Once frozen, changes must be agreed by all.

Item	Value
GATT service UUID	`5781a4a8-0001-41a3-b91f-baa566bff75f`
Data characteristic UUID	`5781a4a8-0002-41a3-b91f-baa566bff75f` (Notify only)
Notification period	2 s (DHT11 max. is about 1 reading/s)
Byte order	Little-endian; frame fits in one notification with the default MTU (20 B of payload)

Offset	Field	Type	Meaning
0	version	uint8	Frame format version (= 1)
1	seq	uint16	Sequence number, used to measure packet loss
3	timestamp	uint32	ms since boot
7	temp	int16	°C × 100 (real)
9	hum	uint16	%RH × 100 (real)
11	water	uint16	mL/min (simulated)
13	posture	uint8	0 = standing, 1 = lying
14	flags	uint8	bit0 crush alarm, bit1 low water, bit2 sensor read failed

Total: 15 bytes. No application CRC: the BLE link layer already adds a CRC-24 to every packet (to be justified in the report).

Dart interface between the BLE layer and the UI:

```
class SensorReading {
  final int seq, timestampMs, waterMlMin;
  final double tempC, humidity;
  final bool lying, crushAlarm, lowWater, sensorError;
}

abstract class ReadingSource {          // two implementations:
  Stream<SensorReading> get readings;   //  - MockSource: generates fake readings in the app
  Stream<bool> get connected;           //  - BleSource: real data from the node
}
```

4. Who does what

Member	Responsibilities	Report section
Lluís Firmware (Device) Has the board	nRF Connect SDK project based on the *peripheral_lbs* sample. DHT11 devicetree overlay and reading. GATT service + notify characteristic. Frame packing. Simulation module (water, buttons with interrupts, alarm logic, flags). LEDs. Integration of the final report.	Device layer, BLE frame spec, report integration
Member 2 Edge BLE layer	Flutter project and repo setup. Android 12+ BLE permissions. Scan filtered by service UUID, connect, enable notifications (CCCD), reconnect. Decoder (bytes to SensorReading) with unit tests on fixed byte arrays. ReadingSource interface, MockSource and BleSource.	Edge layer (BLE), per-layer tests
Member 3 Edge UI	Dashboard screen that only consumes ReadingSource, so it can be built from day 1 with MockSource and no board. Shows: connection status, live values (temp, humidity, water, posture), red alarm banner when the crush flag is set, sensor-error indicator, and a switch between *Simulated* and *Live node*. Stretch goal: line chart of the last readings.	UI description + screenshots for test evidence

Onboarding for Member 3 (day 1): GitHub account, clone the repo, basic Git workflow (pull, commit, push, branch, pull request), Flutter + Android Studio installed and the skeleton app running on a phone. Member 2 does a short pairing session to explain the project structure and MockSource.

5. Timeline

Date	Milestone	Who
22-23 Oct	Freeze contract (UUIDs, frame, Dart interface). Repo + tool setup. Member 3 onboarding.	All
24, 25, 27 Oct	Development of each part (26 Oct: architecture defense, no lab work)	Each owner
28 Oct	Integration session with the board: firmware + app end to end	All, together
29 Oct	Tests (packet loss via seq, reconnection, sensor read failures), screenshots, report sections	All
29 Oct (evening)	Code walk-through: each member explains their part to the other two	All
30 Oct	Submission on Virtual Campus	Lluís

6. Working rules

One GitHub repository with `firmware/`, `app/` and `docs/`. Each member works on their own branch and merges through pull requests. Everyone must be able to explain the whole data path (button press or sensor reading on the board up to the value on the screen), not only their own part.